

$f(t)$ veya $f(x) \rightarrow F(s)$
değişken x veya t 'den s 'ye dönüşür.

$$\mathcal{L}\{f(t)\} = F(s)$$

Laplace formülü: $\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) \cdot e^{-st} dt$

← e'li fonksiyon integrandı.

Ör.

$f(t) = 2$ (sabit fonksiyon)

$$\mathcal{L}\{2\} = \int_0^{\infty} 2 \cdot e^{-st} dt$$
$$= 2 \cdot \int_0^{\infty} e^{-st} = 2 \cdot \frac{e^{-st}}{-s} \Big|_0^{\infty}$$

$$= \frac{2 \cdot e^{-\infty}}{-s} - \left(\frac{2 \cdot e^0}{-s} \right) = 0 - \left(-2/s \right) = \frac{2}{s}$$

$$\int e^{ax+b} dx = \frac{e^{ax+b}}{a} + C$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt \rightarrow \text{genel fonksiyon dönüşümü.}$$

Dönüşüm kuralları:

1) Sabit sayı.

$$\boxed{\mathcal{L}\{a\} = \frac{a}{s}}$$

$$\mathcal{L}\{2\} = 2/s$$

2) $\mathcal{L}\{t\} = \frac{1}{s^2}$ (t yerine x de olabilir).

Yardımcı kurallar: ...

$$4. \mathcal{L}\{a f(t)\} = a \mathcal{L}\{f(t)\}$$

$$\mathcal{L}\{3t\} = 3/s^2$$

$$2. \mathcal{L}\{f(t) \pm g(t)\} = \mathcal{L}\{f(t)\} \pm \mathcal{L}\{g(t)\}$$

$$\mathcal{L}\{t + 5\} = \mathcal{L}\{t\} + \mathcal{L}\{5\} = \frac{1}{s^2} + \frac{5}{s}$$

$$\mathcal{L}\{3t - 4\} = 3/s^2 - 4/s.$$

$$3) \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \quad (n \in \mathbb{Z}^+)$$

$$\mathcal{L}\{t^3\} = \frac{3!}{s^4} = \frac{6}{s^4}, \quad \mathcal{L}\{t^5\} = \frac{120}{s^6}$$

$$\mathcal{L}\{5t^2 - 4t + 3\} = 5 \cdot \frac{2!}{s^3} - \frac{4}{s^2} + \frac{3}{s} = \frac{10}{s^3} - \frac{4}{s^2} + \frac{3}{s}$$

$$4) \mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$\mathcal{L}\{e^t\} = \frac{1}{s-1}, \quad \mathcal{L}\{e^{5t}\} = \frac{1}{s-5}, \quad \mathcal{L}\{e^{-3t}\} = \frac{1}{s+3}$$

$$\mathcal{L}\{2e^{4t}\} = \frac{2}{s-4} //$$

$$5) \mathcal{L}\{\sin(at)\} = \frac{a}{s^2 + a^2}$$

$$\mathcal{L}\{\sin 3t\} = \frac{3}{s^2 + 9}, \quad \mathcal{L}\{\sin t\} = \frac{1}{s^2 + 1}, \quad \mathcal{L}\{\sin 7t\} = \frac{7}{s^2 + 49}$$

$$\mathcal{L}\{3\sin 2t\} = 3 \cdot \frac{2}{s^2 + 4}$$

$$6) \mathcal{L}\{\cos(at)\} = \frac{s}{s^2 + a^2}$$

$$\mathcal{L}\{\cos 2t\} = \frac{s}{s^2 + 4}, \quad \mathcal{L}\{\cos 5t\} = \frac{s}{s^2 + 25}$$

$$\mathcal{L}\{3\cos t\} = 3 \cdot \frac{s}{s^2 + 1} = \frac{3s}{s^2 + 1}.$$

$$7)!!! \mathcal{L}\{e^{at} \cdot t\} = \frac{1}{(s-a)^2} \checkmark$$

e^{at} , yonina geldigi funksiyon

$$L\{e^{at} \cdot t^n\} = \frac{n!}{(s-a)^{n+1}} \checkmark$$

kurulmuş s yerine s-a yazılmasına neden olur.

$$L\{e^{at} \sin(bt)\} = \frac{b}{(s-a)^2 + b^2} \checkmark$$

$$L\{e^{at} \cos(bt)\} = \frac{s-a}{(s-a)^2 + b^2} \checkmark$$

$$L\{e^{2t} \cdot t\} = \frac{1}{(s-2)^2} \checkmark$$

$$L\{e^{-3t} \cdot t^5\} = \frac{120}{(s+3)^6} \checkmark$$

$$L\{e^{4t} \sin 3t\} = \frac{3}{(s-4)^2 + 9} \checkmark$$

$$L\{e^{-5t} \cos 7t\} = \frac{s+5}{(s+5)^2 + 49} \checkmark$$

$$8) L\{t^n \cdot \sin(at)\} = (-1)^n \cdot \frac{d^n}{ds^n} \left(\frac{a}{s^2 + a^2} \right)$$

$$L\{t^n \cdot \cos(at)\} = (-1)^n \cdot \frac{d^n}{ds^n} \left(\frac{s}{s^2 + a^2} \right)$$

$$L\{t \sin 2t\} = (-1)^1 \frac{d}{ds} \left(\frac{2}{s^2 + 4} \right) = -1 \left(\frac{0 \cdot (s^2 + 4) - 4s}{s^4 + 8s^2 + 16} \right) = \frac{4s}{(s^2 + 4)^2}$$

$$L\{t \cos 3t\} = (-1)^1 \frac{d}{ds} \left(\frac{s}{s^2 + 9} \right) = -1 \left(\frac{1 \cdot (s^2 + 9) - 2s^2}{(s^2 + 9)^2} \right) = \frac{s^2 - 9}{(s^2 + 9)^2}$$

TERS LAPLACE DÖNÜŞÜMÜ

$$1. L\{a\} = \frac{a}{s}$$

$$L^{-1}\{1/s\} = 1$$

$$L^{-1}\{3/s\} = 3$$

$$2. L\{t\} = \frac{1}{s^2}$$

$$L^{-1}\{1/s^2\} = t$$

$$L^{-1}\{3/s^3\} = 3t.$$

$$3. L\{t^n\} = \frac{n!}{s^{n+1}}$$

$$L^{-1}\{1/s^3\} = \frac{1}{2} t^2$$

$$L^{-1}\{1/s^5\} = \frac{1}{24} t^4$$

$$L^{-1}\{4/s^6\} = \frac{4}{120} \cdot t^5 = \frac{1}{30} t^5$$

$$4. L\{e^{at}\} = \frac{1}{s-a}$$

$$L^{-1}\left\{\frac{1}{s-2}\right\} = e^{2t}$$

$$L^{-1}\left\{\frac{1}{s+3}\right\} = e^{-3t}$$

$$L^{-1}\left\{\frac{2}{s+5}\right\} = 2e^{-5t}$$

$$5. L\{\sin(at)\} = \frac{a}{s^2+a^2}$$

$$L^{-1}\left\{\frac{1}{s^2+25}\right\} = \frac{\sin 5t}{5}$$

$$L^{-1}\left\{\frac{2}{s^2+4}\right\} = \sin 2t$$

$$L^{-1}\left\{\frac{1}{s^2+49}\right\} = \frac{\sin 7t}{7}$$

$$6. L\{\cos(at)\} = \frac{s}{s^2+a^2}$$

$$L^{-1}\left\{\frac{s}{s^2+1}\right\} = \cos t$$

$$L^{-1}\left\{\frac{s}{s^2+25}\right\} = \cos 5t$$

$$L^{-1}\left\{\frac{s}{s^2+16}\right\} = \cos 4t.$$

$$7. L\{e^{at} \cdot t\} = \frac{1}{(s-a)^2}$$

$$L\{e^{at} \cdot t^n\} = \frac{n!}{(s-a)^{n+1}}$$

$$L\{e^{at} \cdot \sin bt\} = \frac{b}{(s-a)^2 + b^2}$$

$$\left\{ \begin{array}{l} L^{-1}\left\{\frac{1}{(s-3)^2}\right\} = e^{3t} \cdot t \quad \checkmark \\ L^{-1}\left\{\frac{2}{(s+4)^2}\right\} = 2e^{-4t} \cdot t \quad \checkmark \\ L^{-1}\left\{\frac{1}{(s-5)^4}\right\} = \frac{1}{6} \cdot e^{5t} \cdot t^3 \quad \checkmark \end{array} \right.$$

$$\mathcal{L}\{e^{at} \cdot \cos bt\} = \frac{s-a}{(s-a)^2+b^2}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^3}\right\} = \frac{1}{2} e^{-2t} \cdot t^2 \checkmark$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+4)^2+25}\right\} = \frac{1}{5} e^{-4t} \cdot \sin 5t \checkmark$$

$$\mathcal{L}^{-1}\left\{\frac{2}{(s-3)^2+4}\right\} = e^{3t} \cdot \sin 2t \checkmark$$

$$\mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2+16}\right\} = e^{-t} \cdot \cos 4t \checkmark$$

$$\mathcal{L}^{-1}\left\{\frac{s-3}{(s-3)^2+25}\right\} = e^{3t} \cos 5t \checkmark$$

$$\mathcal{L}^{-1}\left\{1/s^5\right\} = \frac{t^4}{24} \checkmark$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-3)^4}\right\} = e^{3t} \cdot \frac{t^3}{6} \checkmark$$

$$\mathcal{L}^{-1}\{4/s\} = 4 \checkmark$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s+5}\right\} = e^{-5t} \checkmark$$

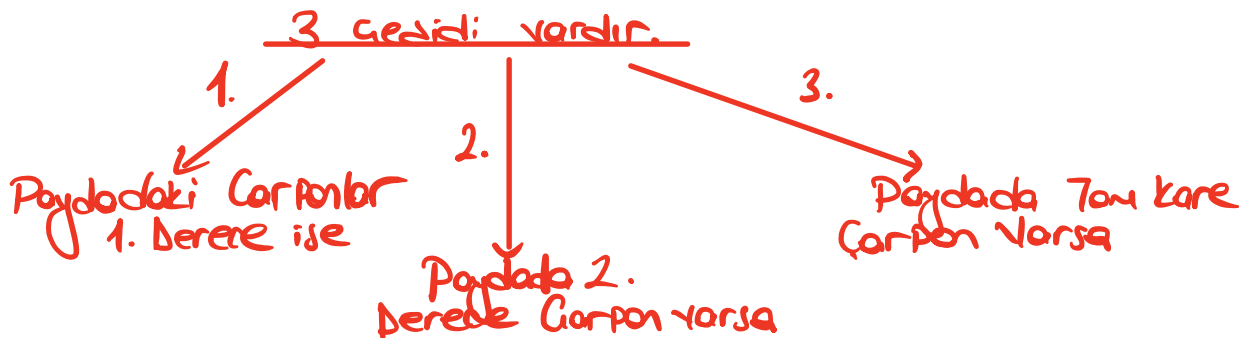
$$\mathcal{L}^{-1}\left\{\frac{6}{s^2+9}\right\} = 2 \sin 3t \checkmark$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+49}\right\} = \cos 7t \checkmark$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2+81}\right\} = e^{2t} \cdot \frac{\sin 9t}{9} \checkmark$$

$$\mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+25}\right\} = e^{-2t} \cdot \cos 5t \checkmark$$

Basit kesirlere Ayırma (Partial fraction decomposition) gerektiren Ters Laplace Dönüşümü.



1. Geşit: Paydadaki Carponlar 1. Derere
 $(s+2), (s-3) \dots$

Ex. $\mathcal{L}^{-1} \left\{ \frac{2s-3}{s^2+3s+2} \right\}$

$$\frac{2s-3}{(s+2)(s+1)} = \frac{A}{s+2} + \frac{B}{s+1} = \frac{7}{s+2} - \frac{5}{s+1}$$

$\begin{matrix} \nearrow 7 \\ \downarrow -2 \end{matrix} \quad \begin{matrix} \nearrow -5 \\ \downarrow -1 \end{matrix}$

$$\mathcal{L}^{-1} \left\{ \frac{7}{s+2} - \frac{5}{s+1} \right\} = \underbrace{\mathcal{L}^{-1} \left\{ \frac{7}{s+2} \right\}}_{7e^{-2t}} - \underbrace{\mathcal{L}^{-1} \left\{ \frac{5}{s+1} \right\}}_{5e^{-t}} = 7e^{-2t} - 5e^{-t}$$

Ex. $\mathcal{L}^{-1} \left\{ \frac{1}{s^2-4} \right\} = \mathcal{L}^{-1} \left\{ \frac{1}{(s-2)(s+2)} \right\}$

e^{at} den dolayı $s \rightarrow s-a \Rightarrow$ translation property.

$$\frac{1}{(s-2)(s+2)} = \frac{A}{s-2} + \frac{B}{s+2} = \frac{1/4}{s-2} - \frac{1/4}{s+2}$$

$\begin{matrix} \nearrow 1/4 \\ \downarrow 2 \end{matrix} \quad \begin{matrix} \nearrow -1/4 \\ \downarrow -2 \end{matrix}$

$$\Rightarrow \mathcal{L}^{-1} \left\{ \frac{1}{(s-2)(s+2)} \right\} = \underbrace{\mathcal{L}^{-1} \left\{ \frac{1/4}{s-2} \right\}}_{e^{2t} \cdot 1/4} - \underbrace{\mathcal{L}^{-1} \left\{ \frac{1/4}{s+2} \right\}}_{e^{-2t} \cdot 1/4} = \frac{1}{4} (e^{2t} - e^{-2t})$$

Pozitif, negatif olarak bütün çarpanlar 1. Dereceye getiriliyor
 $\Rightarrow$ Partial Fraction Decomposition.

2. Çeşit: Pozitif 2. Dereceden Çarpan Varsa:

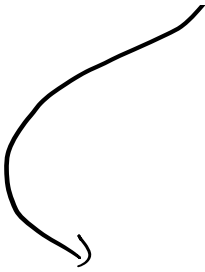
$(s^2+4), (s^2+9), \rightarrow$ Çarpanlarına ayrılıp 2. Derece kollar.

$\mathcal{L}^{-1} \left\{ \frac{2s+1}{(s-2)(s^2+9)} \right\}$

Bilinen bir kural değil.

$$\frac{2s+1}{(s-2)(s^2+9)} = \frac{A}{s-2} + \frac{Bs+C}{s^2+9}$$

$$2s+1 = As^2+9A + Bs^2+Cs - 2Bs-2C$$



$$(A+B)s^2 = 0 \Rightarrow A+B = 0$$

$$(C-2B)s = 2s \Rightarrow C-2B=2$$

$$8A-2C=1$$

$$C+2A=2 \Rightarrow \begin{array}{r} 8A-2C=1 \\ 4A+2C=4 \\ \hline 13A=5 \Rightarrow A=5/13 \end{array}$$

$$B = -5/13$$

$$C = 2 - 10/13 = 6/13$$

Parçaladıktan sonra ayrı ayrı Laplace T. alıp kuralları uyguluyoruz.

$$C = 2 - 10/13 = 6/13$$

3. Çeşit: Payda da Tam Kare Görünür Varsa

$$s^2, (s-1)^2, (s+3)^2, \dots$$

$$\mathcal{L}^{-1} \left\{ \frac{2s-3}{(s+4)(s-2)^2} \right\}$$

kural değil.

$$\frac{2s-3}{(s+4)(s-2)^2} = \frac{A}{s+4} + \frac{B}{s-2} + \frac{C}{(s-2)^2}$$

$$\Rightarrow A(s^2 - 4s + 4) + B(s^2 + 2s - 8) + C(s+4) = 2s-3$$

$$= A s^2 + B s^2 - 4A s + 2B s + C s + 4A - 8B + 4C = 2s-3$$

$$A+B=0 \Rightarrow B=-A$$

$$-4A+2B+C = 6B+C=2 \Rightarrow 24B+4C=8$$

$$4A-8B+4C = -12B+4C=3 \Rightarrow 12B-4C=3$$

$$36B=11 \Rightarrow B=11/36, A=-11/36$$

$$\frac{11}{3} - 4C = 3 \Rightarrow 4C = \frac{2}{3} \Rightarrow C = \frac{2}{12} = \frac{1}{6}$$

$$\Rightarrow \mathcal{L}^{-1} \left\{ \frac{2s-3}{(s+4)(s-2)^2} \right\} = \mathcal{L}^{-1} \left\{ \frac{-11/36}{s+4} + \frac{11/36}{s-2} + \frac{1/6}{(s-2)^2} \right\}$$

$$= \underbrace{-\frac{11}{36} \mathcal{L}^{-1}\left\{\frac{1}{s+4}\right\}}_{-\frac{11}{36} e^{-4t}} + \underbrace{\frac{11}{36} \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\}}_{\frac{11}{36} e^{2t}} + \underbrace{\frac{1}{6} \mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2}\right\}}_{\frac{1}{6} e^{2t} \cdot t}$$

-Tonkoreye Tonoulara -

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2+2s+2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2+1}\right\} = \sin t \cdot e^{-t}$$

$$1. \mathcal{L}^{-1}\left\{\frac{1}{s^2+6s+13}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{(s+3)^2+4}\right\} = \frac{1}{2} e^{-3t} \sin 2t$$

$$2. \mathcal{L}^{-1}\left\{\frac{s+2}{s^2+4s+5}\right\} = \mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+1}\right\} = e^{-2t} \cos t.$$

$$3. \mathcal{L}^{-1}\left\{\frac{1}{s^2-4s+13}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2+9}\right\} = \frac{1}{3} e^{2t} \sin 3t$$

Laplace ile Diferansiyel Denklemleri Çözme.

1. Soru: Initial Value Problem olmalı.

$$\underbrace{y(0)=\dots \quad y'(0)=\dots}_{0 \text{ noktasında olmalı.}}$$

$y(1)=a, y'(1)=b$ Laplace ile çözülemez.

Ör: $y'' - 3y' - 4y = 0 \rightarrow y(0)$ ve $y'(0)$ bilinmelidir.

$y' - 4y = t + 3 \rightarrow y(0)$ bilinse yeterlidir.

...

2. Soru: y ve y' li ifadelerin katsayıları sabit olmalıdır.

~~$t y'' - 3t y' + y = 0$~~ Çözülmez!

$\vec{t}$ veya x 'e bağlı fonksiyon.
 $\mathcal{L}\{y\} = Y(s)$

$$\mathcal{L}\{y'\} = sY(s) - y(0)$$

$$\mathcal{L}\{y''\} = s^2 Y(s) - sy(0) - y'(0)$$

Soru: $y' - 2y = e^{2t}$, $y(0) = 1$.

1. adım:

$$\mathcal{L}\{y'\} - 2\mathcal{L}\{y\} = \mathcal{L}\{e^{2t}\}$$

$$sY(s) - y(0) - 2Y(s) = \frac{1}{s-2} \quad \text{amaç } Y(s)'i \text{ yalnız bırakmak.}$$

$$(s-2)Y(s) - y(0) = \frac{1}{s-2} \Rightarrow Y(s) = \frac{1}{(s-2)^2} + \frac{y(0)}{s-2} \rightarrow 1$$

$$\Rightarrow Y(s) = \frac{s-1}{(s-2)^2}$$

2. adım: Laplace cinsinden çözümü bulduk, şimdi ters Laplace olarak esas fonksiyonu buluyoruz.

$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{s-1}{(s-2)^2}\right\} (*)$$

$$\frac{s-1}{(s-2)^2} = \frac{A}{(s-2)} + \frac{B}{(s-2)^2} \Rightarrow As - 2A + B = 2s - 1$$
$$\Rightarrow A = 1, B - 2 = -1 \Rightarrow B = 1$$

rewriting (*) using this, we obtain

$$\mathcal{L}^{-1}\left\{\frac{s-1}{(s-2)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2}\right\} = e^{2t} + e^{2t} \cdot t = y(t)$$

Soru:

$$y'' - 2y' - 3y = 0, \quad y(0) = 0, \quad y'(0) = 2.$$

$$\begin{aligned} 1. \quad \mathcal{L}\{y''\} - 2\mathcal{L}\{y'\} - 3\mathcal{L}\{y\} &= s^2 \cancel{y(0)} - \cancel{s} y'(0) - y'(0) - 2(s \cancel{y(0)} - \cancel{y(0)}) - 3y(s) \\ &= s^2 y(s) - 2 - 2s y(s) - 3y(s) = y(s)(s^2 - 2s - 3) - 2 = 0 \\ &\Rightarrow y(s) = \frac{2}{(s^2 - 2s - 3)} \end{aligned}$$

$$2. \quad \mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{(s^2 - 2s - 3)}\right\} \quad (*)$$

$$\frac{2}{(s^2 - 2s - 3)} = \frac{A}{(s-3)} + \frac{B}{(s+1)} \Rightarrow As + Bs + A - 3B = 2$$

$$A = -B, \quad 4A = 2 \Rightarrow A = 1/2, \quad B = -1/2$$

rewriting (*) using this, we have

$$\mathcal{L}^{-1}\{y(s)\} = 1/2 \mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} - 1/2 \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} = \frac{e^{3t}}{2} - \frac{e^{-t}}{2} = y(t).$$

Soru: $y'' - 4y = 2e^{3t}, \quad y(0) = 0, \quad y'(0) = 0$

$$1. \quad \mathcal{L}\{y''\} - 4\mathcal{L}\{y\} = 2\mathcal{L}\{e^{3t}\}$$

$$\Rightarrow s^2 y(s) - s y(0) - y'(0) - 4y(s) = \frac{2}{s-3}$$

$$\Rightarrow (s^2 - 4)y(s) = \frac{2}{s-3} \Rightarrow y(s) = \frac{2}{(s-3)(s^2 - 4)}$$

$$2. \quad \mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{(s-3)(s^2 - 4)}\right\} \quad (*)$$

$$\frac{2}{(s-3)(s^2 - 4)} = \frac{2}{(s-3)(s-2)(s+2)} = \frac{A}{(s-3)} + \frac{B}{(s-2)} + \frac{C}{(s+2)}$$

$$(s^2 - 4) \quad (s^2 - 5s + 6) \quad (s^2 - 5s + 6)$$

$$\Rightarrow As^2 + Bs^2 + Cs^2 - Bs - 5Cs - 4A - 6B + 6C = 2.$$

$$\Rightarrow A+B+C=0 \Rightarrow A - 4C=0 \quad -16C+30C+6C=2 \Rightarrow 20C=2 \Rightarrow C=\frac{1}{10}$$

$$B+5C=0$$

$$\downarrow$$

$$-5C$$

$$B = -\frac{1}{2}$$

$$A = \frac{+2}{5}$$

rewriting (*) using this we obtain

$$\mathcal{L}^{-1}\{Y(s)\} = \frac{2}{5} \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} - \frac{1}{2} \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} + \frac{1}{10} \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\}$$

$$\Rightarrow y(t) = \frac{2}{5} e^{1t} - \frac{1}{2} e^{2t} + \frac{1}{10} e^{-2t}$$

$$x'' + 5x' + 4x = e^t, \quad x(0) = 1, \quad x'(0) = -1.$$

$$\text{Let } X(s) = \mathcal{L}\{x(t)\}$$

taking L.T of the eqn:

$$\mathcal{L}\{x''\} + 5\mathcal{L}\{x'\} + 4\mathcal{L}\{x\} = s^2 X(s) - sx(0) - x'(0) + 5(sX(s) - x(0)) + \frac{4}{s}$$

$$= \cancel{s^2 X(s)} - s + 1 + \cancel{5sX(s)} - 5 + \frac{4}{s} = \frac{1}{s-1}$$

$$\Rightarrow (s^2 + 5s) X(s) - s + \frac{4}{s} - 4 = \frac{1}{s-1}$$

$$\Rightarrow (s^2 + 5s) X(s) = \frac{s^2 - 4}{s} + \frac{1}{s-1} + 4 \Rightarrow \frac{s^2 - 4s - s^2 + 4s + s + 4s^2 - 4}{s^2 - s} = X(s)(s^2 + 5s)$$

$$\Rightarrow \frac{s^3 + 3s^2 - 3s}{s^2 - s} = \frac{s^2 + 3s - 3}{s-1} = (s^2 + 5s) X(s) \Rightarrow X(s) = \frac{s^2 + 3s - 3}{(s-1)(s^2 + 5s + 4)}$$

$$\Rightarrow X(s) = \frac{s^2 + 3s - 3}{(s-1)(s+1)(s+4)} = \frac{A}{(s-1)} + \frac{B}{(s+1)} + \frac{C}{(s+4)}$$

$$\Rightarrow A(s^2 + 5s + 4) + B(s^2 + 5s - 4) + C(s^2 - 1) = s^2 + 3s - 3$$

$$\Rightarrow As^2 + Bs^2 + Cs^2 = s^2 \Rightarrow A+B+C=1$$

$$5As + 3Bs = 3s \Rightarrow 5A+3B=3$$

$$4A - 4B - C = -3 \quad \frac{1}{2} \quad 3B = \frac{5}{2} \Rightarrow B = \frac{5}{6}$$

$$5A - 3B = -2$$

$$10A = 1 \Rightarrow A = 1/10 \quad \frac{6}{60} + \frac{50}{60} + C = \frac{60}{60} \Rightarrow C = \frac{4}{60} = \frac{1}{15}$$

Then, taking the inverse L.T:

$$\mathcal{L}^{-1}\{X(s)\} = \frac{1}{10} e^t + \frac{5}{6} e^{-t} + \frac{1}{15} e^{-4t} = x(t)$$

Konvolüsyon integrali

$$f(t) * g(t) = \int_0^t f(t-T) \cdot g(T) dT \rightarrow \text{ilk fonksiyonu } T \text{ kadar sağa kaydırıp ikinci fonksiyonun } T \text{ deki değeriyle carp.}$$

$$t * t^2 = \int_0^t (t-T) \cdot T^2 dT = \int_0^t tT^2 - T^3 dT = \left(\frac{tT^3}{3} - \frac{T^4}{4} \right) \Big|_0^t$$

$$= \frac{t^4}{3} - \frac{t^4}{4} = \frac{t^4}{12}$$

Konvolüsyon-Laplace ilişkisi

$$\mathcal{L}\{f(t) * g(t)\} = F(s) \cdot G(s) \Rightarrow f(t) * g(t) = \mathcal{L}^{-1}\{F(s) \cdot G(s)\}$$

Konvolüsyon işleminin Laplace dönüşümü carpma işlemine denktir.

$$\mathcal{L}\{t^2 * e^{3t}\} = \frac{2}{s^3} \cdot \frac{1}{s-3}$$

$$\mathcal{L}\{t^3 * \cos 3t\} = \frac{6}{s^4} \cdot \frac{s}{s^2+9}$$

$$\mathcal{L}\left\{\int_0^t (t-u)^2 \sin 2u \, du\right\} = \mathcal{L}\{t^2 * \sin 2t\} = \mathcal{L}\{t^2\} \cdot \mathcal{L}\{\sin 2t\} \\ = \frac{2}{s^3} \cdot \frac{2}{s^2+4} //$$

$$\mathcal{L}\left\{\int_0^t e^{3(t-\tau)} \cdot \cos \tau \, d\tau\right\} = \mathcal{L}\{e^{3t} * \cos t\} = \mathcal{L}\{e^{3t}\} \cdot \mathcal{L}\{\cos t\} \\ = \frac{1}{s-3} \cdot \frac{s}{s^2+1}$$

Carpma Durumundaki s' lere konvolüsyon integrali ile ters laplace dönüşümü uygulayabiliriz.

find $\mathcal{L}^{-1}\left\{\frac{1}{s^3+4s}\right\}$ using convolution.

$$\mathcal{L}^{-1}\left\{\frac{1}{s^3+4s}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s} \cdot \frac{1}{s^2+4}\right\} = 1 * \frac{\sin 2t}{2}$$

$\begin{array}{cc} \uparrow & \uparrow \\ 1 & \frac{\sin 2t}{2} \end{array}$

$$1 * \frac{\sin 2t}{2} = \int_0^t \frac{\sin 2u}{2} \, du = \frac{\sin 2t}{2} * 1 = \int_0^t \frac{\sin 2t-u}{2} \, du \quad ???$$

$$\underbrace{-\frac{1}{2} \frac{\cos 2u}{2}} \bigg|_0^t = -\frac{\cos 2t}{4} - \underbrace{\left(-\frac{\cos 0}{4}\right)}_{-\frac{1}{4}} = \frac{-\cos 2t + 1}{4}$$

Diferansiyelin içinde integral varsa mutlaka konvolüsyon soruyordur ve Laplace'ten başka çözümü yoktur.

Or.

$$y' - 3y = \int_0^t (t-\tau) e^{\tau} d\tau, \quad y(0) = 0$$

$$\underbrace{\mathcal{L}\{y'\} - 3\mathcal{L}\{y\}}_{(s-3)Y(s)} = \underbrace{\mathcal{L}\left\{\int_0^t (t-\tau) e^{\tau} d\tau\right\}}_{\mathcal{L}\{t * e^t\}}$$

$$(s-3)Y(s).$$

$$\mathcal{L}\{t * e^t\} = \frac{1}{s^2} \cdot \frac{1}{s-1}$$

Burdon Devanuke.